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Effect of film morphology on oxygen and water interaction with copper phthalocyanine

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Abstract

Copper phthalocyanine (CuPc) films of thickness 25 nm and 100 nm were grown by thermal sublimation at 25°C, 150°C, and 250°C in order to vary morphology. Using a source-measure unit and a quartz crystal microbalance (QCM), we measured changes in electrical resistance and film mass in situ during exposure to controlled pulses of O_2 and H_2O vapor. Mass loading by O_2 was enhanced by a factor of 5 in films deposited at 250°C, possibly due to the ~200°C CuPc α→β transition which allows higher O_2 mobility between stacked molecules. While gas/vapor sorption occurred over timescales of < 10 minutes, resistance change occurred over timescales > 1 hour, suggesting that mass change occurs by rapid adsorption at active surface sites, whereas resistive response is dominated by slow diffusion of adsorbates into the film bulk. Resistive response generally increases with film deposition temperature due to increased porosity associated with larger crystalline domains. The 25 nm thick films exhibit higher resistive response than 100 nm thick films after an hour of O_2/H_2O exposure due to the smaller analyte diffusion length required for reaching the film/electrode interface. We found evidence of decoupling of CuPc from the gold-coated QCM crystal due to preferential adsorption of O_2/H_2O molecules on gold, which is consistent with findings of other studies.

Introduction

Emergence of wearable electronics and other smart textile technologies highlights the importance of low-cost functional materials for flexible thin film devices[1]. Copper phthalocyanine (CuPc) is an aromatic macrocycle being explored for application in flexible photovoltaics (OPVs)[2], light-emitting diodes (OLEDs)[3], transistors (OTFTs)[4], biosensors[5], and gas sensors[6]. Its p-type behavior, strong gas response, absorption in the UV range, and low cost due to industrial-scale production (> 60,000 tons annually[7]) as a blue pigment for inks, plastics, and textiles makes CuPc well-suited as an active material in emerging wearable technologies[8,9].

Although CuPc is thermally and chemically stable under ambient condition[10], adsorption of gas molecules can facilitate charge transfer which compromises the intrinsic electronic properties of CuPc film[11]. Electronic conductivity of metal phthalocyanines arises primarily from doping by electron-accepting O_2 and H_2O, which increases hole density and alters surface potential by addition of negatively-charged adsorbate ions to the film surface[12,13]. Oxygen coordination occurs at the central metal atom[14] while H_2O sorption is generally attributed to hydrogen bonding near domain boundaries[15,16]. Upon exposure to water, formation of H_2O multi-layers on CuPc domain surfaces may lead to H_2O dissociation and conduction of ions which further increases conductivity[17,18].

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Moreover, O_2/H_2O adsorption may facilitate fracture, displacement, and rearrangement of CuPc domains by disturbing intermolecular and molecule-substrate adhesion[19].

The response of metal phthalocyanines to organic vapors and noxious gases has been well-characterized[16,19,20]. It is recognized that film morphology and thickness play a role in CuPc gas response but mechanisms for their effects are still not fully understood[21-23]. During thermal deposition of CuPc, molecules align by weak van der Waals and π-π interactions, resulting in electron delocalization along the Cu atom stacking axis[9-10]. Substrate heating enhances mobility of deposited molecules, allowing formation of longer molecular chains which aggregate to form elongated crystallites. Crystalline domain growth is often associated with changes in film porosity and roughness[17,24]. Gas response time decreases and mass loading is enhanced with film porosity due to increased surface area[15,25]. Surface area and number of inter-grain boundaries which impede charge transport typically decrease as domains grow larger[26]. Since response time is primarily related to slow diffusion of adsorbates into the film bulk[11], response time is inversely related to film thickness since surface area to volume ratio decreases as thickness increases[10]. As film thickness decreases, charge transport layers lie closer to the film/gas interface and therefore the effects of adsorption-induced surface trap states is enhance[20].

An understanding of the influence of film morphology on CuPc gas response is critical for extending device lifetimes and improving gas sensor performance. Here we study the CuPc gas/vapor response by measuring changes in electrical resistance and mass of films exposed to O_2 and H_2O. This allows us to observe the relationship between gas sorption/desorption kinetics and the electronic gas/vapor response.

Experiment

Purified CuPc was purchased from Sigma Aldrich. CuPc films were deposited by thermal sublimation onto pre-cleaned platinum interdigitated electrodes and gold-plated 5 MHz AT-cut quartz crystal microbalance (QCM) crystals. Evaporation was carried out in a Nanomaster thermal evaporator. During thermal evaporation, substrates were held at 25°C, or heated to 150°C, and 250°C. Film thickness was monitored in situ using a QCM system to produce both 25 nm and 100 nm thick films at each deposition temperature.

Prior to gas/vapor sorption measurements, films were placed under 10^{-6} Torr (1.33×10^{-6} mbar) vacuum for 3 hours. QCM frequency change was recorded using an SRS QCM200 and electrical resistance was recorded using a Keithley 6430 source-meter with preamplifier. Pulses of O_2 and H_2O vapor with pressures 5, 10, 15, and 20 Torr were injected into the vacuum chamber using two mass flow controllers. A LabVIEW program was used to control the gas/vapor flow and record resistivity and frequency shift during the experiment. All measurements were performed at 27 ± 0.5°C using a modified pressure/flow control system from Surface Measurement Systems. Frequency shift of the QCM crystals during the experiment was converted to mass change using the well-known Sauerbrey equation[27].

To confirm that morphology of the CuPc films changed with deposition temperature, atomic force microscope (AFM) images of each 25 nm thick film were acquired using a NT-MDT NTEGRA Spectra microscope. The AFM was operated in semi-contact (tapping) mode at 73 kHz tip resonance frequency with a line scan rate of 0.7 Hz.

Results and Discussion

AFM images of 25 nm thick films deposited at 25°C, 150°C, and 250°C are shown in Figure 1. Film morphology is consistent with temperature-induced grain size growth observed in other studies on CuPc films[26]. As temperature increases, domains become larger and more elongated as molecules stack along the central Cu atom axis. Films deposited at 250°C exhibit a significant 3-fold increase in roughness over films deposited at 25°C and 150°C. As discussed later, this is likely associated with higher porosity.

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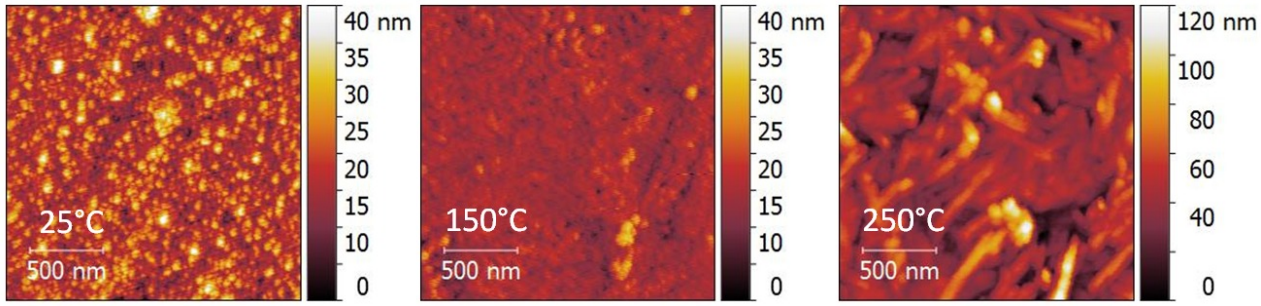

Figure 1. AFM images of CuPc films deposited at 25°C, 150°C, and 250°C.

Resistance response (R/R_0) and mass change (ΔM) of CuPc films exposed to different pressures of H_2O vapor and O_2 are shown in Figure 2 during 4-pulse sequences of H_2O and O_2 (shown in bottom panel). Resistance decreases in response to both H_2O and O_2, indicating that both analytes behave as electron acceptors. H_2O/O_2 sorption leads to charge transfer which increases hole density, resulting in higher conductivity. This result supports the findings of other studies[12,13]. In films deposited at temperatures over 25°C, resistance changes occur over a long timescale (> 2 hours), even during isobaric conditions. Thus, not all films reach steady-state conditions before the next gas pulse begins. During the 5 Torr H_2O pulse, R/R_0 and ΔM kinetics are qualitatively similar to those of the next pulses. This suggests that a lack of equilibrium before subsequent gas pulses does not significantly change the nature of the resistive gas response. Since the focus of this experiment is to study the effect of morphology on gas response, and each film is exposed to the same sequence of gas pulses, it is informative to compare responses between films even if full equilibrium has not been achieved.

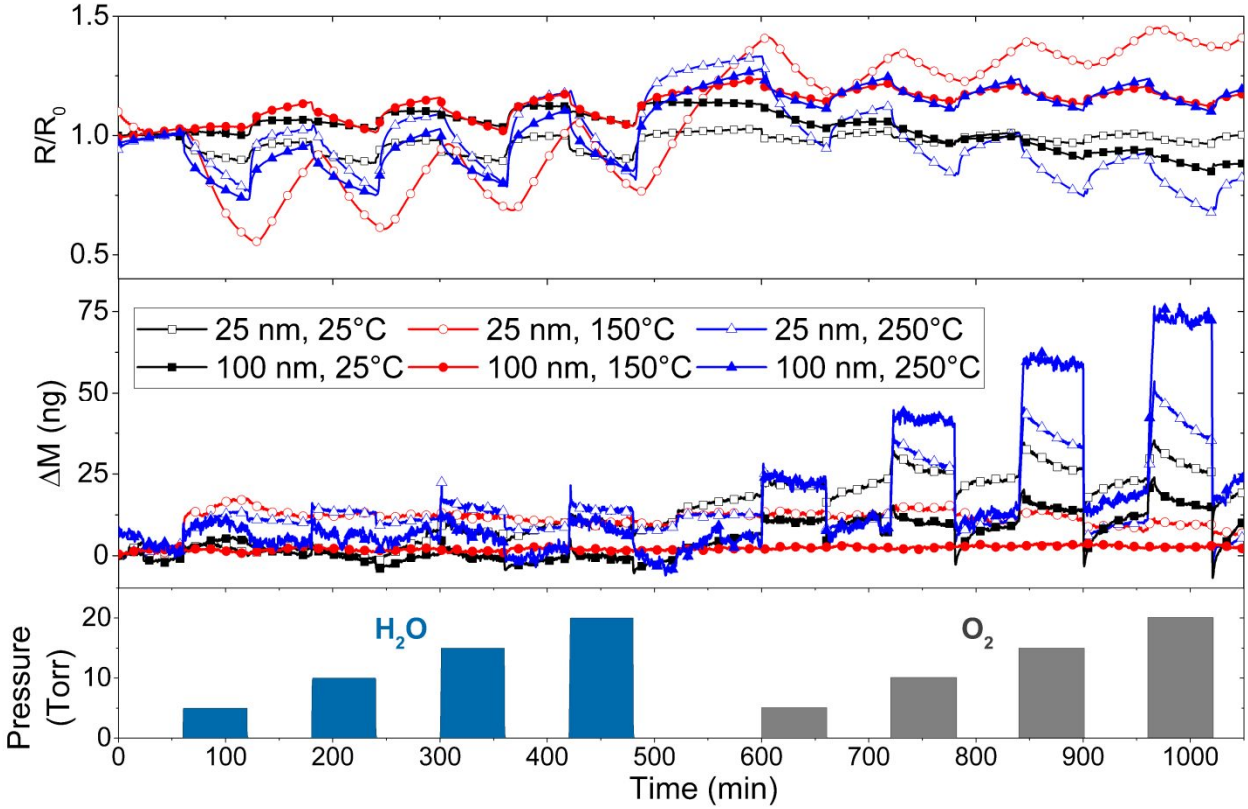

Figure 2. Resistance response (R/R_0, top panel) and mass change (ΔM, middle panel) of CuPc films during exposure to different pressures (bottom panel) of H_2O vapor (blue pulses) and O_2 (gray pulses). Films are designated by their thickness (25 or 100 nm) and the temperature of the substrate during deposition (25°C, 150°C, or 250°C).

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As shown in the middle panel of Figure 2, mass loading of films occurs most significantly during O_2 exposure and is monotonic with pressure. The apparent decrease in mass of some films during the gas/vapor pulses is more closely examined in Figure 5b. In order to better study effects of morphology on the gas/vapor response, Figure 3 shows normalized and background-corrected R/R_0 and ΔM for films exposed to 20 Torr H_2O and O_2 flows for 1 hour. The 20 Torr gas/vapor responses are analyzed here because the responses are stronger than at lower analyte pressures.

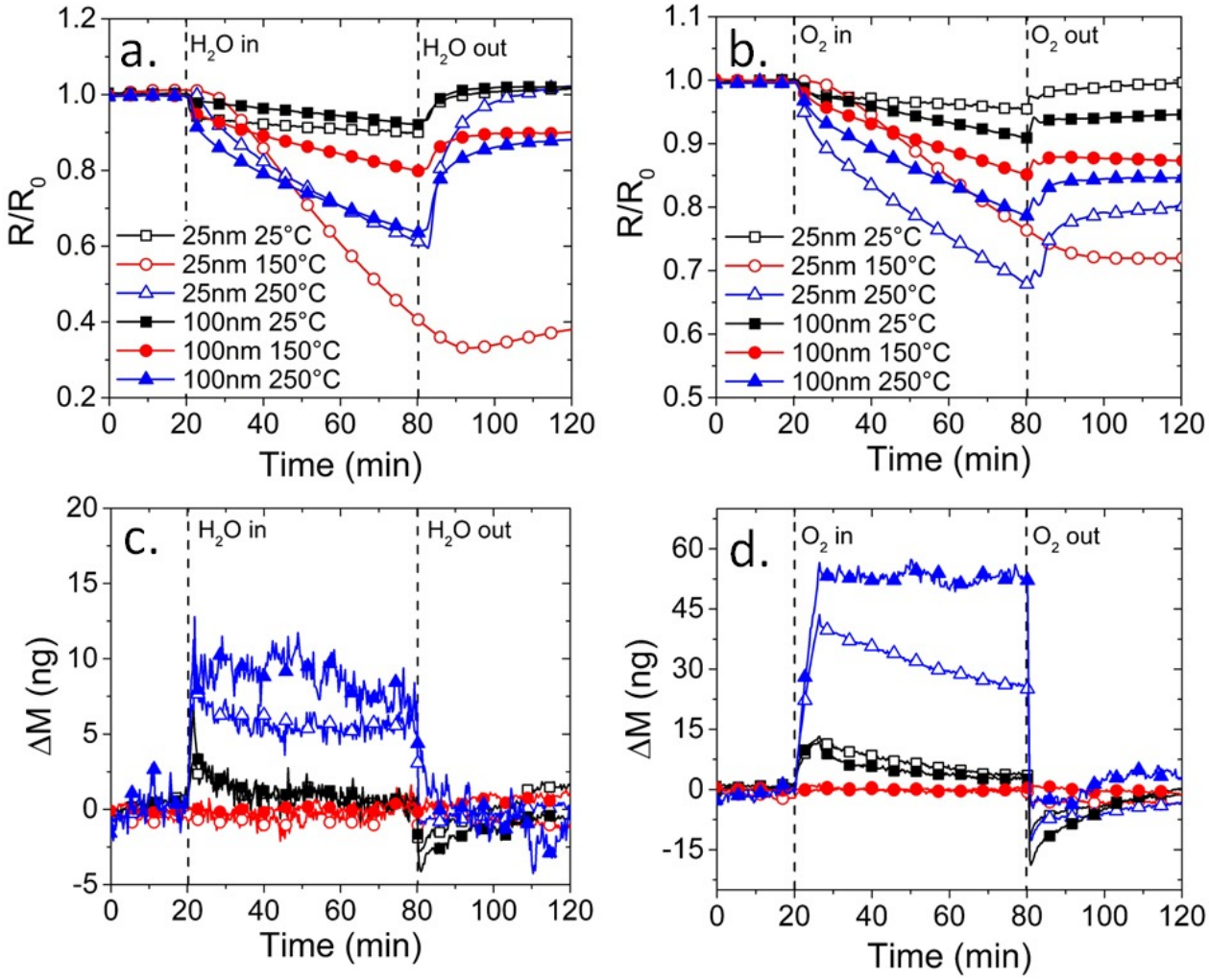

Figure 3. Resistance change (R/R_0) during exposure to 20 Torr pulses of H_2O vapor (a) and O_2 (b) for 1 hour. Mass change (ΔM) during exposure to 20 Torr H_2O vapor (c) and O_2 (d) for 1 hour.

Figure 3 shows the relationship between R/R_0 and ΔM responses during gas/vapor exposure. Figure 3a and 3b show R/R_0 during introduction and removal of H_2O and O_2 (marked by vertical dashed lines). Corresponding changes in mass during the same gas/vapor pulses are shown in Figures 3c and 3d. Upon initial exposure to H_2O/O_2, rapid mass loading occurs, primarily in films deposited at 250°C. The change in mass ΔM is larger for 100 nm films than 25 nm films, which is expected since thicker films necessarily contain more active adsorption sites than thinner films. Desorption of H_2O/O_2 occurs less than 5 minutes after films are placed back under vacuum. ΔM appears almost completely reversible. In contrast, changes in resistance occur over time scales of > 1 hour and exhibit poor reversibility, especially for O_2 desorption. The difference in time scales associated with R/R_0 and ΔM responses suggests that different processes are responsible for resistance and mass changes in the films. Mass loading on film

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surfaces and domain boundaries occurs rapidly but does not significantly alter charge carrier density near the charge transport layers. Since films were deposited on patterned electrodes, it is likely that only charge carriers in domains near the film-substrate interface contribute to the resistive gas/vapor resistance. Only after adsorbates diffuse deep enough into the film bulk to reach charge transport regions near the CuPc-electrode interface, which occurs on the order of hours, does conductivity increase. Kerp et al. reports that mass loading saturation of phthalocyanine films by O_2 adsorption occurs on a timescale of minutes[18]. That data reported here suggest that the saturation reported by Kerp is a consequence of adsorbate crowding at surface adsorption sites, not saturation of the bulk film by O_2, which appears to take hours. The slow diffusion of adsorbates toward the film-electrode interface observed here is consistent with the findings of Passard et al.[11].

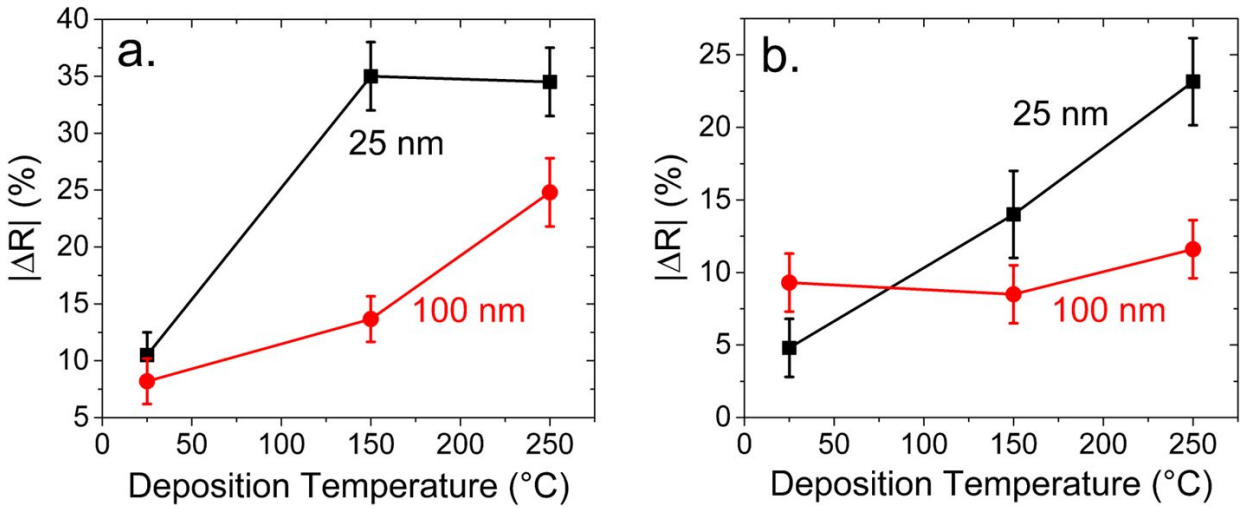

Figure 4. Resistance change (ΔR) of 25 nm and 100 nm thick films during exposure to 20 Torr pulses of H_2O vapor (a) and O_2 (b) as a function of film deposition temperature.

Figure 4 shows magnitude of the resistance change (ΔR) of each film after 1 hour exposure to 20 Torr pulses of H_2O vapor (a) and O_2 (b). The ΔR response to both H_2O and O_2 generally increases with film deposition temperature. This is likely due to increases in film porosity which result from the growth of larger domains. The porosity improves access to active adsorption sites by facilitating adsorbate diffusion through open channels, increasing the rate at which adsorbates reach charge transport layers near the film/electrode interface. ΔR is generally higher for 25 nm films than 100 nm films since adsorbate migration to electrodes occurs more rapidly. This finding agrees with the results of others who show that film thickness is directly related to the magnitude and response time of the resistive CuPc gas/vapor response[11,25].

Mass changes in films during 20 Torr gas/vapor pulses are shown in Figure 5a. As mentioned previously, mass loading occurs most significantly in films deposited at 250°C. Mass loading of O_2 in particular is enhanced by over a factor of 5 in films deposited at 250°C. It is possible that this is a consequence of the CuPc α→β transition. At roughly 200°C, CuPc undergoes a meta-stable triclinic α-phase to stable monoclinic β-phase transition characterized by an increase in the tilting angle between the molecular and stacking axes, resulting in less π-π overlap between adjacent molecules[10]. The transition increases spacing between the central metal atoms of adjacent stacked molecules from roughly 3.8 Å to 5 Å[28]. Since the van der Waals radius of O_2 is approximately 1.5 Å[29], extra space between β-phase stacked molecules may allow greater O_2 mobility toward central Cu atoms. O_2 coordination with CuPc occurs mainly at Cu atoms[14], so the formation of β-phase domains may significantly enhance O_2 coordination and result in the clear increase in mass loading by O_2 in films deposited at 250°C. The CuPc α→β transition is also associated with an increase in film roughness[30] which leads to higher surface area and an increase in the number of accessible adsorption sites. This effect may be responsible for the slight increase in H_2O loading of films deposited at 250°C. Since H_2O adsorption occurs primarily around grain boundaries[15,16], it is expected that the α→β transition has less of an effect on H_2O loading than on O_2 loading. The 100 nm thick films exhibit higher ΔM than 25 nm thick

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films because they necessarily contain more adsorption sites and thus allow accommodation of more adsorbate molecules.

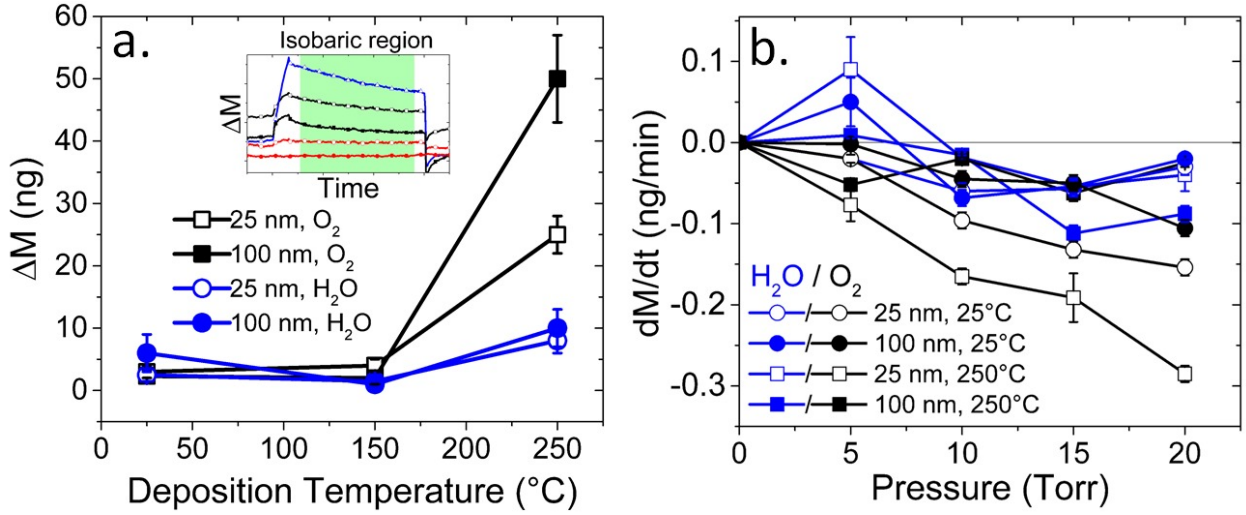

Figure 5. a. Total mass change (ΔM) in response to 20 Torr pulses of O_2 (black) and H_2O (blue), shown for 25 nm and 100 nm thick films deposited at 25°C, 150°C, and 250°C. The inset is related to Figure 5b. b. Average time derivative of film mass (dM/dt) during O_2 exposure (black) and H_2O exposure (blue) after initial rapid mass loading has occured. dM/dt is extracted during O_2/H_2O exposure under isobaric conditions, highlighted in green in the inset of (a). For clarity, data from films deposited at 150°C are not shown since dM/dt ≈ 0 g/s.

It is informative to examine the behavior of ΔM during H_2O/O_2 exposure. The average time derivative of film mass (dM/dt) was extracted from ΔM curves shown in Figure 2 during H_2O/O_2 loading. dM/dt values were averaged during isobaric conditions, shown highlighted in green in the inset of Figure 5a as an example. Continuous H_2O/O_2 loading of rigid films would result in dM/dt > 0 during the entire length of the gas/vapor pulse. However, after an initial mass increase which occurs on a time scale of ~5 minutes, we observe that dM/dt < 0 in all films except for those exposed to H_2O at 5 Torr. Since it is unlikely that significant desorption of adsorbates occurs during isobaric conditions, we propose that sorption-induced changes in mechano-structural and/or viscoelastic properties of the films are responsible for the apparent decreases in mass. Voinova et al. showed that materials undergoing changes in viscoelasticity may display signatures of mass loss when deposited on QCM[31]. This is related to breakdown of the Sauerbrey equation, which is strictly valid only for thin uniform films which are rigidly bound to the quartz oscillator[27]. The apparent loss of mass has been observed in other organic films undergoing adsorption of H_2O and O_2 on QCM crystals[32]. Since H_2O sorption generally occurs by hydrogen bonding near grain boundaries[15,16] and can result in the formation of an H_2O meniscus on the CuPc surface[17,18], it is likely that high H_2O pressures enhance aggregation of H_2O near inter-grain interfaces. This increases the probability of slippage between adjacent grains as they oscillate at the 5 MHz QCM resonant frequency f_0. Inter-grain slip and CuPc-gold slip increase the resonant frequency of the oscillator, which results in the apparent decrease in film mass. Ojifinni et al. find that H_2O adsorption on gold surfaces may result in formation of H_2O-Au, H_2O-O, or OH complexes when oxygen adatoms are present[33], and Park et al. find evidence of CuPc molecule displacement from gold substrates due to preferential O_2/H_2O adsorption on gold[19]. Both of these processes represent pathways for CuPc-gold decoupling which increases f_0 as CuPc molecules are effectively lifted off the QCM surface. The positive value of dM/dt during H_2O loading at 5 Torr provides evidence that the limited water adsorption at this pressure does not induce significant mechano-structural changes in the films.

The mass rate dM/dt exhibits higher pressure dependence during O_2 loading than during H_2O loading. O_2 coordination with central Cu atoms[14] may disrupt π-π stacking and lead to domain fracture and rearrangement of CuPc[19]. Preferential adsorption of O_2 on gold may be responsible for displacing CuPc from the QCM surface, leading to an increase in f_0. The O_2-induced decrease in mass is most evident in 25 nm thick films, which show a roughly linear decrease in dM/dt with O_2 pressure. Slow diffusion of O_2 through the CuPc[11] to the QCM surface must occur before disruption of CuPc-gold adhesion takes place, so it is reasonable that dM/dt observed in 25 nm

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films is more pronounced than in 100 nm films. The pressure dependence of dM/dt is a result of faster diffusion of O_2 into the bulk film and to the film-crystal interface when more adsorbate molecules are present.

Conclusions

We investigated separately effects of film morphology and thickness on the interaction of H_2O and O_2 with CuPc. By measuring electrical resistance of films and frequency shift of CuPc-coated QCM crystals, we studied the relation between sorption/desorption kinetics and film conductivity. While mass loading occurred relatively quickly after gas/vapor exposure (~10 minutes), resistance changes occurred over much longer time scales (> 1 hour). This suggests that the resistive CuPc gas/vapor response is dominated by slow diffusion of adsorbates into the film bulk, while mass change occurs by rapid adsorption at active surface sites. Resistive response generally increases with film deposition temperature due to increased porosity associated with larger domains. 25 nm thick films exhibit higher resistive response than 100 nm thick films due to the smaller analyte diffusion length required for reaching the film/electrode interface. O_2 loading of films was enhanced by a factor of 5 in films deposited at 250°C. We suggest that this may be a consequence of the CuPc $\alpha \rightarrow \beta$ phase transition which occurs around 200°C and results in higher O_2 mobility between stacked molecules, facilitating O_2 coordination at the central Cu atoms. Finally, we found evidence of H_2O/O_2-induced mechano-structural changes in the films. This is likely due to slip between CuPc molecules, slip between CuPc and the QCM oscillator, and to decoupling of CuPc from the gold-coated QCM crystal caused by preferential adsorption of H_2O/O_2 molecules on gold.

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References

1. Hu, J., Meng, H., Li, G. and Ibekwe, S.I., 2012. A review of stimuli-responsive polymers for smart textile applications. Smart Materials and Structures, 21(5), p.053001. http://dx.doi.org/10.1088/0964-1726/21/5/053001

2. Tang, C.W., 1986. Two-layer organic photovoltaic cell. Applied Physics Letters, 48(2), pp.183-185. DIO: 10.1063/1.96937

3. Tadayyon, S.M., Grandin, H.M., Griffiths, K., Norton, P.R., Aziz, H. and Popovic, Z.D., 2004. CuPc buffer layer role in OLED performance: a study of the interfacial band energies. Organic Electronics, 5(4), pp.157-166. doi:10.1016/j.orgel.2003.10.001

4. Royer, J.E., Park, J., Colesniuc, C., Lee, J.S., Gredig, T., Lee, S., Jin, S., Schuller, I.K., Trogler, W.C. and Kummel, A.C., 2010. Mobility saturation in tapered edge bottom contact copper phthalocyanine thin film transistors.Journal of Vacuum Science & Technology B, 28(4), pp.C5F22-C5F27.

5. Sotomayor, M.D.P.T., Tanaka, A.A. and Kubota, L.T., 2002. Development of an enzymeless biosensor for the determination of phenolic compounds. Analytica Chimica Acta, 455(2), pp.215-223. doi:10.1016/S0003-2670(01)01608-7

6. Wang, X., Ji, S., Wang, H. and Yan, D., 2011. Highly sensitive gas sensor enhanced by tuning the surface potential. Organic Electronics, 12(12), pp.2230-2235. doi:10.1016/j.orgel.2011.09.014

7. Kimura, M. and Shirai, H., 2003. The Porphyrin Handbook-Applications of Phthalocyanines Vol. 19. Elsevier.

8. Farooq, A., Karimov, K.S., Ahmed, N., Ali, T., Alamgir, M.K. and Usman, M., 2015. Copper phthalocyanine and metal free phthalocyanine bulk heterojunction photodetector. Physica B: Condensed Matter, 457, pp.17-21. doi:10.1016/j.physb.2014.09.032

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9. de la Torre, G., Claessens, C.G. and Torres, T., 2007. Phthalocyanines: old dyes, new materials. Putting color in nanotechnology. Chemical Communications, (20), pp.2000-2015. DOI: 10.1039/b614234f

10. Strelcov, E. and Kolmakov, A., 2008. Copper phthalocyanine quasi-1D nanostructures: growth morphologies and gas sensing properties. Journal of nanoscience and nanotechnology, 8(1), pp.212-221. DOI: http://dx.doi.org/10.1166/jnn.2008.N11

11. Passard, M., Pauly, A., Blanc, J.P., Dogo, S., Germain, J.P. and Maleysson, C., 1994. Doping mechanisms of phthalocyanines by oxidizing gases: application to gas sensors. Thin Solid Films, 237(1), pp.272-276. doi:10.1016/0040-6090(94)90272-0

12. Zhivkov, I., Nešpůrek, S. and Schauer, F., 1999. Influence of oxygen on the parameters of a thin film copper phthalocyanine field effect transistor.Advanced Materials for Optics and Electronics, 9(5), pp.175-180. DOI: 10.1002/1099-0712

13. Yan, X., Wang, H. and Yan, D., 2006. An investigation on air stability of copper phthalocyanine-based organic thin-film transistors and device encapsulation. Thin Solid Films, 515(4), pp.2655-2658. doi:10.1016/j.tsf.2006.05.040

14. Bohrer, F.I., Colesniuc, C.N., Park, J., Ruidiaz, M.E., Schuller, I.K., Kummel, A.C. and Trogler, W.C., 2008. Comparative gas sensing in cobalt, nickel, copper, zinc, and metal-free phthalocyanine chemiresistors. Journal of the American Chemical Society, 131(2), pp.478-485. DOI: 10.1021/ja803531r

15. Park, J., Royer, J.E., Colesniuc, C.N., Bohrer, F.I., Sharoni, A., Jin, S., Schuller, I.K., Trogler, W.C. and Kummel, A.C., 2009. Ambient induced degradation and chemically activated recovery in copper phthalocyanine thin film transistors. Journal of Applied Physics, 106(3), p.034505. http://dx.doi.org/10.1063/1.3159885

16. Bohrer, F.I., Sharoni, A., Colesniuc, C., Park, J., Schuller, I.K., Kummel, A.C. and Trogler, W.C., 2007. Gas sensing mechanism in chemiresistive cobalt and metal-free phthalocyanine thin films. Journal of the American Chemical Society, 129(17), pp.5640-5646. DOI: 10.1021/ja0689379

17. Aziz, F., Sayyad, M.H., Sulaiman, K., Majlis, B.H., Karimov, K.S., Ahmad, Z. and Sugandi, G., 2012. Influence of humidity conditions on the capacitive and resistive response of an Al/VOPc/Pt co-planar humidity sensor. Measurement Science and Technology, 23(1), p.014001. http://dx.doi.org/10.1088/0957-0233/23/1/014001

18. Kerp, H.R., Westerduin, K.T., Van Veen, A.T. and Van Faassen, E.E., 2001. Quantification and effects of molecular oxygen and water in zinc phthalocyanine layers. Journal of Materials Research, 16(02), pp.503-511. DOI: http://dx.doi.org/10.1557/JMR.2001.0073

19. Park, J.H., Lee, S. and Kummel, A.C., 2015. Domain fracture and recovery process of metal phthalocyanine monolayers via NO2 and H2O. Journal of Vacuum Science & Technology B, 33(3), p.030604. http://dx.doi.org/10.1116/1.4919227

20. Yang, R.D., Gredig, T., Colesniuc, C.N., Park, J., Schuller, I.K., Trogler, W.C. and Kummel, A.C., 2007. Ultrathin organic transistors for chemical sensing. Applied physics letters, 90(26), p.263506. http://dx.doi.org/10.1063/1.2749092

21. Hsiao, Y.S., Whang, W.T., Suen, S.C., Shiu, J.Y. and Chen, C.P., 2008. Morphological control of CuPc and its application in organic solar cells. Nanotechnology, 19(41), p.415603. http://dx.doi.org/10.1088/0957-4484/19/41/415603

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22. Armand, F., Ferez, H., Fouriaux, S., Araspin, O., Pradeau, J.P., Claessens, C.G., Maya, E.M., Vazquez, P. and Torres, T., 1999. Tetraamidometallo-phthalocyanines Langmuir-Blodgett Films: Morphology versus central metal effects on NO2 detection. Synthetic metals, 102(1), pp.1476-1477. doi:10.1016/S0379-6779(98)00555-4

23. Gao, J., Xu, J.B., Zhu, M., Ke, N. and Ma, D., 2007. Thickness dependence of mobility in CuPc thin film on amorphous SiO2 substrate. Journal of Physics D: Applied Physics, 40(18), p.5666. http://dx.doi.org/10.1088/0022-3727/40/18/022

24. Xiao, K., Liu, Y., Yu, G. and Zhu, D., 2003. Influence of the substrate temperature during deposition on film characteristics of copper phthalocyanine and field-effect transistor properties. Applied Physics A, 77(3-4), pp.367-370. DOI: 10.1007/s00339-003-2169-6

25. Hsieh, J.C., Liu, C.J. and Ju, Y.H., 1998. Response characteristics of lead phthalocyanine gas sensor: effects of film thickness and crystal morphology. Thin Solid Films, 322(1), pp.98-103. doi:10.1016/S0040-6090(97)00964-4

26. Gentry, K.P., Gredig, T. and Schuller, I.K., 2009. Asymmetric grain distribution in phthalocyanine thin films. Physical Review B, 80(17), p.174118. http://dx.doi.org/10.1103/PhysRevB.80.174118

27. Sauerbrey, G., 1959. Verwendung von Schwingquarzen zur Wägung dünner Schichten und zur Mikrowägung. Zeitschrift für physik, 155(2), pp.206-222. DOI 10.1007/BF01337937

28. Yoo, H.K., Kee, C.S., Kang, C., Hwang, I.W. and Lee, J.W., 2013. Characteristics of carrier injection on polarization-dependent photoexcitation in copper phthalocyanine thin films. Journal of Nanophotonics, 7(1), pp.073795-073795. http://dx.doi.org/10.1117/1.JNP.7.073795

29. Bondi, A., 1964. van der Waals volumes and radii. The Journal of physical chemistry, 68(3), pp.441-451. DOI: 10.1021/j100785a001

30. Gredig, T., Gentry, K.P., Colesniuc, C.N. and Schuller, I.K., 2010. Control of magnetic properties in metallo-organic thin films. Journal of materials science, 45(18), pp.5032-5035. DOI: 10.1007/s10853-010-4319-5

31. Voinova, M.V., Jonson, M. and Kasemo, B., 2002. 'Missing mass' effect in biosensor's QCM applications. Biosensors and Bioelectronics, 17(10), pp.835-841. doi:10.1016/S0956-5663(02)00050-7

32. Muckley, E.S., Nelson, A.J., Jacobs, C.B. and Ivanov, I.N., 2016. Multimodal probing of oxygen and water interaction with metallic and semiconducting carbon nanotube networks under ultraviolet irradiation. Journal of Photonics for Energy, 6(2), pp.025506-025506. doi:10.1117/1.JPE.6.025506

33. Ojifinni, R.A., Froemming, N.S., Gong, J., Pan, M., Kim, T.S., White, J.M., Henkelman, G. and Mullins, C.B., 2008. Water-enhanced low-temperature CO oxidation and isotope effects on atomic oxygen-covered Au (111). Journal of the American Chemical Society, 130(21), pp.6801-6812. DOI: 10.1021/ja800351j

34. Surface Measurement Systems Ltd. Unit 5, Wharfside Rosemont Road, Alperton London, HA0 4PE United Kingdom.

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